α-labeling of banana tree graphs

Borna Gojšić, Anamari Nakić

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Abstract

A banana tree graph, denoted by $B(n,k)$, is a tree constructed by connecting n star graphs S_k to a root vertex. This article builds upon previous research on graceful labelings of banana trees and explores their α-labelability. We classify α-labelings for $B(n,1)$ and $B(1,k)$ trees. We present α-labelings of the $B(n,k)$ tree, for $k \geq \frac{n}{2}+1$ and $n \in \mathbb{N}$. Finally, we prove that the $B(n,k)$ tree is not α-labelable for $1 < k \leq \frac{n}{2}$ and any $n \geq 3$.

Keywords: graph labeling; graceful labeling; α-labeling; banana tree graph

1 Introduction

An interesting problem of graceful labeling or β-labeling of a graph was first introduced by Rosa in [6]. A *tree* is a graph in which any two vertices are connected by exactly one path. The question of whether a given tree allows a graceful labeling turned out to be a hard problem. A great open problem in graph theory is the Ringel-Kotzig conjecture that all trees are graceful. This topic has been a focus of many papers, an overview of many results can be found in famous dynamic survey by Gallian [2].

In the same paper [6], an α-labeling (or a α-valuation) was also defined as a graceful labeling with an additional property. Trees with α-labelings have proven to be useful in the development of the theory of graph decompositions. Known results on this topic can also be found in [2] where open questions are also stated.

Banana tree $B(n,k)$, is a hierarchical tree constructed by connecting n star graphs S_k to a central vertex of the tree. They were introduced in [1]. It has been shown that banana trees and extended banana trees admit graceful labeling [7, 8].

~~Summary of Graceful Results G bananas [2566], [2565] Chen, Lu, and Yeh [639] define a firecracker as a graph obtained from the concatenation of stars by linking one leaf from each. They also define a banana tree as a graph obtained by connecting a vertex v to one leaf of each of any number of stars (v is not in any of the stars). They proved that firecrackers are graceful and conjecture that banana trees are graceful. Before Sethuraman and Jesintha [2566] and [2565] (see also [1242]) proved that all banana trees and extended banana trees (graphs obtained by joining a vertex to one leaf of each of any number of stars by a path of length of at least two) are graceful, various kinds of bananas trees had been shown to be graceful by Bhat-Nayak and Deshmukh [508], by Murugan and Arumugam [2020], [2018] and by Vilfred [3093].~~

However, the more restrictive nature of the α-labelings of a tree motivates the question of whether all banana trees possess this property. This paper investigates the

α-labelability of banana trees, presenting both theoretical and computational results. We provide explicit constructions for α-labelings in certain cases, formulate conjectures regarding their existence for specific parameter ranges, and show non-existence results. With this paper we hope to contribute to the the study of graph labelings. Some of our generalization results were motivated by the work in [4].

We will first show that all $B(n, 1)$ trees are α-labelable for all $n \in \mathbb{N}$ as well as all $B(1, k)$ trees. We will then show that all $B(n, k)$ trees are α-labelable for all $n \in \mathbb{N}$ and $k \geq \frac{n}{2} + 1$. Finally, we will show that all $B(n, k)$ trees are not α-labelable for all $1 < k \leq \frac{n}{2}$ and $n \geq 3$.

2 Basic notions and results

Definition 1. A *graph* $G = (V, E)$ is a pair of a nonempty finite set V of vertices and a finite set E of edges. An edge is an unordered pair of two vertices.

An edge $e = \{v, w\}$ will be briefly denoted by $e = vw$ and we will say that the vertices v and w are *adjacent* or *neighboring*. Furthermore, we say that the edge e is *incident* to v and w. We define the mapping $\deg_G : V(G) \to \mathbb{N}$ so that $\deg_G(v)$ is the number of adjacent vertices of v in G, called the *degree* of v. A *leaf* is a vertex of degree 1.

A *subgraph* H of a graph G is a graph such that $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$. We say that H is an *induced subgraph* of G if $E(H)$ consist of all of the edges from $E(G)$ that connect pairs of vertices in $V(H)$. A *path* P of a graph G is an array of edges of the following form $v_1v_2, v_2v_3, \ldots, v_{n-1}v_n$. The *length of a path* P is the number of edges. A path P of length $n-1$ in a graph G can be viewed as a subgraph of G, and it is called *a path graph* P_n. We say that a graph is *connected* if there exists a path between any of its two vertices.

Definition 2. A *tree* is a graph in which any two vertices are connected by exactly one path. An equivalent definition is that a tree with n vertices is a connected graph with $n - 1$ edges.

Let us denote the set of integers from 1 to n by $[n] := \{1, 2, \ldots, n\}$.

Definition 3. Let $G = (V, E)$ be a graph with n vertices and m edges. A *labeling* of G is a mapping $f : V \to \mathbb{N}$. We say that a labeling is *graceful* if $\operatorname{Im} f \subseteq [m + 1]$, the induced edge labeling $f^* : E \to \mathbb{N}$ defined by

$$f^*(vw) = |f(v) - f(w)|$$

is injective and

$$\operatorname{Im} f^* = \{f^*(vw) : vw \in E\} = [n - 1].$$

A labeling of G is an *α-labeling* if it is graceful and if there exists a constant $\alpha \in \mathbb{N}$ such that

$$f(v) \leq \alpha < f(w) \text{ or } f(w) \leq \alpha < f(v), \text{ for all } vw \in E.$$

Other names for α-labelings are *balanced*, *interlaced*, and *strongly graceful*.

A graph G for which there exists an α-labeling is said to be *α-labelable*. We say that two labelings f and g are *isomorphic* if there exists an automorphism $\sigma : V \to V$ of G such that $f = g \circ \sigma$. Two labelings are *distinct* if and only if they are not isomorphic.

Remark 1. [5]

1. If f is an α-labeling of G and $f^*(vw) = 1$ for some $v, w \in V$, then $\alpha = \min\{f(v), f(w)\}$.
2. If f is an α-labeling of a graph G with n vertices, then its complementary labeling $\overline{f}$ defined as $\overline{f}(v) = n + 1 - f(v)$ is also an α-labeling of G with $\overline{\alpha} = n - \alpha + 1$.
3. If f is an α-labeling of G, its reverse (or inverse) labeling f^r defined as

$$f^r(v) = \begin{cases} \alpha + 1 - f(v), & f(v) \leq \alpha \\ n + \alpha + 1 - f(v), & f(v) > \alpha \end{cases}$$

is also an α-labeling of G with $\alpha^r = \alpha$.

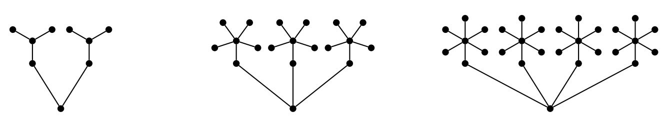
A star S_k is the complete bipartite graph $K_{1,k}$, a tree with one internal node, *the center* of the star, and k leaves.

S_6 S_7

Figure 1: Examples of star trees

Definition 4. A *banana tree* $B(n, k)$ is a tree obtained by joining a central vertex of the tree, *the root*, with one leaf of each of the n copies of a star graph S_k.

From the definition for $k = 1$, we can see that $B(n, 1) \cong K_{1,n} \cong S_n$. Examples of $B(n, k)$ for $k > 2$ trees are shown in Figure 2.



$B(2, 4)$ $B(3, 6)$ $B(4, 7)$

Figure 2: Examples of $B(n, k)$ trees

2.1 Rooted Standard Form of a Banana Tree

A *rooted standard form* of a $B(n, k)$ tree is a way of representing the tree by specifically naming the vertices. This form of representing the $B(n, k)$ tree is motivated by its symmetries and the results on α-labelings we obtained. A general $B(n, k)$ tree in this form is shown in Figure 3.

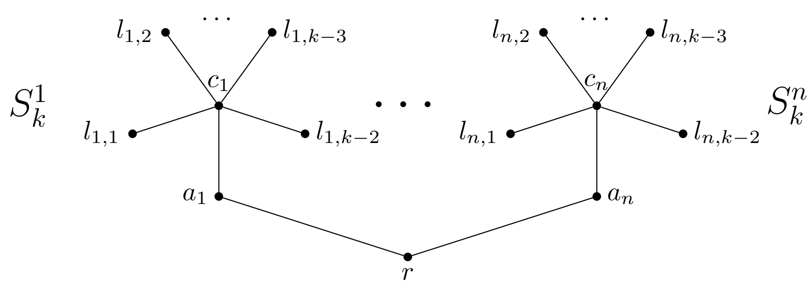


Figure 3: Rooted standard form of a $B(n,k)$ tree

In this representation the vertex set of a $B(n,k)$ tree for $k > 1$ is

$$V(B(n,k)) = \{r\} \cup \{a_i \mid i \in [n]\} \cup \{c_i \mid i \in [n]\} \cup \{l_{i,j} \mid i \in [n], j \in [k-2]\}.$$

The vertex we named r is the root of $B(n,k)$, the vertices a_i are adjacent to the root and c_i are the centers of the stars S_k^i, the induced i-th star subgraph of $B(n,k)$. To improve the readability of the paper later on, we introduce the following notation for two subsets of vertices of S_k^i : $\mathcal{L}_i = \{l_{i,j} \mid j \in [k-2]\}$ and $\mathcal{B}_i = \{a_i\} \cup \mathcal{L}_i$. We remark that swapping the labelings of two stars S_k^i and $S_k^{i'}$ does not change the labeling of $B(n,k)$. Also, swapping the labels of any two leaves $l_{i,j}$ and $l_{i,j'}$ of S_k^i does not change the labeling of $B(n,k)$. The edge set is

$$E(B(n,k)) = \{ra_i \mid i \in [n]\} \cup \{a_ic_i \mid i \in [n]\} \cup \{c_il_{i,j} \mid i \in [n], j \in [k-2]\}.$$

When $k = 1$, the vertex set is

$$V(B(n,1)) = \{r\} \cup \{a_i \mid i \in [n]\}$$

and the edge set is

$$E(B(n,1)) = \{ra_i \mid i \in [n]\}.$$

The $B(n,k)$ tree has $nk+1$ vertices and nk edges.

3 α-labelings of a $B(n,k)$ tree

Let $f : V \to \mathbb{N}$ be a labeling of a $B(n,k)$ tree. For an arbitrary set of vertices S of $B(n,k)$ tree we will write

$$\mathcal{F}(S) = \operatorname{Im} f\big|_S = \{f(x) \mid x \in S\}.$$

Also, we will write just $\mathcal{F}$ when we are looking at values of f across the whole domain, i.e. $\mathcal{F} = \operatorname{Im} f$.

Definition 5. Let $\mathcal{A}(n,k)$ be the set of all α-labelings of the $B(n,k)$ tree and $\mathcal{A}_x(n,k)$ be the set of all α-labelings of the $B(n,k)$ tree where $f(r) = x$. Also, we will use the notation $A(n,k) = |\mathcal{A}(n,k)|$ and $A_x(n,k) = |\mathcal{A}_x(n,k)|$.

Proposition 1. *Let f be an α-labeling of the $B(n,k)$ tree for any $n \in \mathbb{N}$ and $k \geq 2$. Then,*

$$f(r) \leq \alpha \iff f(c_i) \leq \alpha \quad \forall i \in [n]$$

Proof. (⇐): Let $f(c_i) \leq \alpha$ for all $i \in [n]$. Then, since f is an α-labeling and $c_i a_i$ is an edge of the $B(n,k)$ tree for all $i \in [n]$, we have $f(c_i) \leq \alpha < f(a_i)$ for all $i \in [n]$. Furthermore, since ra_i is an edge of the $B(n,k)$ tree for all $i \in [n]$, we must have $f(r) \leq \alpha < f(a_i)$, and thus we have proven one direction of the equivalence.

(⇒): Let $f(r) \leq \alpha$. Then, we must have $f(r) \leq \alpha < f(a_i)$ for all $i \in [n]$, since f is an α-labeling and ra_i is an edge of the $B(n,k)$ tree for all $i \in [n]$. Furthermore, we must have $f(c_i) \leq \alpha < f(a_i)$ for all $i \in [n]$, since f is an α-labeling and $c_i a_i$ is an edge of the $B(n,k)$ tree for all $i \in [n]$. ☐

Theorem 1. *Let f be an α-labeling of the $B(n,k)$ tree for any $n \in \mathbb{N}$ and $k \geq 2$. Then, exactly one of the following holds:*

1. *$\mathcal{F}(RC) = [n+1]$ and $\alpha = n+1$, or*
2. *$\mathcal{F}(RC) = \{nk+1, nk, \ldots, nk+1-n\}$ and $\alpha = nk-n$,*

where $RC = \{r\} \cup \{c_i \mid i \in [n]\}$.

Proof. By Proposition 1 we have that

$$f(r) \leq \alpha \iff f(c_i) \leq \alpha \quad \forall i \in [n]$$

Therefore, we will look at the two cases: when $f(r) \leq \alpha$ and when $f(r) > \alpha$.

1. Let $f(r) \leq \alpha$. Then, we have $f(c_i) \leq \alpha$ for all $i \in [n]$. We must have $f(v) > \alpha$ for all $v \in V \setminus RC$, since for every $v \in V \setminus RC$, vc_i is an edge of $B(n,k)$ for some $i \in [n]$. Thus, we have exactly $|RC| = n+1$ numbers that are $\leq \alpha$. Therefore, we have $\alpha = n+1$.
2. Let $f(r) > \alpha$. Then, we have $f(c_i) > \alpha$ for all $i \in [n]$. We must have $f(v) \leq \alpha$ for all $v \in V \setminus RC$, since for every $v \in V \setminus RC$, vc_i is an edge of $B(n,k)$ for some $i \in [n]$. Thus, we have exactly $|V \setminus RC| = nk+1-n+1 = nk-n$ numbers that are $\leq \alpha$. Therefore, we have $\alpha = nk-n$.

☐

Theorem 2. *For every $n, k \in \mathbb{N}$, we have*

$$\overline{\mathcal{A}_x}(n,k) = \mathcal{A}_{nk+2-x}(n,k),$$

where $\overline{\mathcal{A}_x}(n,k) = \{\overline{f} : f \in \mathcal{A}_x(n,k)\}$

Proof. Let $f \in \mathcal{A}_x(n,k)$ be an α-labeling of the $B(n,k)$ tree with $f(r) = x$. Then, by Remark 1, we have that $\overline{f}(v) = n+1-f(v)$ for all $v \in V(B(n,k))$ and it is an α-labeling. Thus, we have $\overline{f}(r) = nk+2-x$ and therefore,

$$\overline{\mathcal{A}_x}(n,k) \subseteq \mathcal{A}_{nk+2-x}(n,k)$$

Analogously, let $g \in \mathcal{A}_{nk+2-x}(n,k)$ be an α-labeling of the $B(n,k)$ tree with $g(r) = nk+2-x$. Then, we have that $\overline{g}(v) = n+1-g(v)$ for all $v \in V(B(n,k))$. Thus, we have $\overline{g}(r) = x$ and therefore,

$$\mathcal{A}_{nk+2-x}(n,k) \subseteq \overline{\mathcal{A}_x}(n,k)$$

Finally, we have that $\overline{\mathcal{A}_x}(n,k) = \mathcal{A}_{nk+2-x}(n,k)$. ☐

Corollary 2.1. *For every $n, k \in \mathbb{N}$, we have*

$$\mathcal{A}(n,k) = \bigcup_{x=1}^{n+1} (\mathcal{A}_x(n,k) \cup \overline{\mathcal{A}_x}(n,k))$$

Proof. Since we have that $\mathcal{F}(RC)$ is either $[n+1]$ or $\{nk+1, nk, \ldots, nk+1-n\}$, we know that the following holds:

$$\mathcal{A}(n,k) = \left(\bigcup_{x=1}^{n+1} \mathcal{A}_x(n,k) \right) \cup \left(\bigcup_{x=nk+1-n}^{nk+1} \mathcal{A}_x(n,k) \right)$$

Then, by Theorem 2, we have

$$\begin{aligned} \mathcal{A}(n,k) &= \left(\bigcup_{x=1}^{n+1} \mathcal{A}_x(n,k) \right) \cup \left(\bigcup_{x=1}^{n+1} \overline{\mathcal{A}_x}(n,k) \right) \\ &= \bigcup_{x=1}^{n+1} (\mathcal{A}_x(n,k) \cup \overline{\mathcal{A}_x}(n,k)) \end{aligned}$$

□

Corollary 2.2. *For every $n \in \mathbb{N}$ and $k \geq 2$, we have*

$$A(n,k) = 2 \sum_{x=1}^{n+1} A_x(n,k)$$

Proof. By the previous corollary, we have that

$$\begin{aligned} A(n,k) = |\mathcal{A}(n,k)| &= \left| \bigcup_{x=1}^{n+1} (\mathcal{A}_x(n,k) \cup \overline{\mathcal{A}_x}(n,k)) \right| \\ &= \sum_{x=1}^{n+1} |\mathcal{A}_x(n,k)| + |\overline{\mathcal{A}_x}(n,k)| = 2 \sum_{x=1}^{n+1} A_x(n,k) \end{aligned}$$

□

Theorem 3. *For every $n, k \in \mathbb{N}$ and $x \in [n+1]$, we have*

$$\mathcal{A}_x^r(n,k) = \mathcal{A}_{n+2-x}(n,k),$$

where $\mathcal{A}_x^r(n,k) = \{f^r : f \in \mathcal{A}_x(n,k)\}$.

Proof. Let $f \in \mathcal{A}_x(n,k)$ be an α-labeling of the $B(n,k)$ tree with $f(r) = x \in [n+1]$. Then, by Remark 1, we have that $f^r(v) = n+2-f(v)$ for all $v \in RC$, since $\alpha = n+1$. Thus, we have $f^r(r) = n+2-x$ and therefore,

$$\mathcal{A}_x^r(n,k) \subseteq \mathcal{A}_{n+2-x}(n,k).$$

Analogously, let $g \in \mathcal{A}_{n+2-x}(n,k)$ be an α-labeling of the $B(n,k)$ tree with $g(r) = n+2-x$. Then, we have that $g^r(v) = n+2-g(v)$ for all $v \in RC$. Thus, we have $g^r(r) = x$ and therefore,

$$\mathcal{A}_{n+2-x}(n,k) \subseteq \mathcal{A}_x^r(n,k).$$

Finally, we have that $\mathcal{A}_x^r(n,k) = \mathcal{A}_{n+2-x}(n,k)$. □

Corollary 3.1. *For every $n, k \in \mathbb{N}$, we have*

$$\mathcal{A}(n,k) = \bigcup_{x=1}^{\lfloor \frac{n}{2} \rfloor + 1} \left(\mathcal{A}_x(n,k) \cup \mathcal{A}_x^r(n,k) \cup \overline{\mathcal{A}_x}(n,k) \cup \overline{\mathcal{A}_x^r}(n,k) \right).$$

Proof. We obviously have that:

$$\bigcup_{x=1}^{n+1} \mathcal{A}_x(n,k) = \left(\bigcup_{x=1}^{\lfloor \frac{n}{2} \rfloor + 1} \mathcal{A}_x(n,k) \right) \cup \left(\bigcup_{x=\lfloor \frac{n}{2} \rfloor + 2}^{n+1} \mathcal{A}_x(n,k) \right)$$

By the previous corollary, we have that

$$\begin{aligned}\bigcup_{x=1}^{n+1} \mathcal{A}_x(n,k) &= \left(\bigcup_{x=1}^{\lfloor \frac{n}{2} \rfloor + 1} \mathcal{A}_x(n,k) \right) \cup \left(\bigcup_{x=1}^{\lfloor \frac{n}{2} \rfloor + 1} \mathcal{A}_x^r(n,k) \right) \\ &= \bigcup_{x=1}^{\lfloor \frac{n}{2} \rfloor + 1} \left(\mathcal{A}_x(n,k) \cup \mathcal{A}_x^r(n,k) \right).\end{aligned}$$

By Theorem 2, we have that

$$\bigcup_{x=1}^{n+1} \overline{\mathcal{A}_x}(n,k) = \bigcup_{x=1}^{\lfloor \frac{n}{2} \rfloor + 1} \left(\overline{\mathcal{A}_x}(n,k) \cup \overline{\mathcal{A}_x^r}(n,k) \right).$$

Finally, we have that

$$\begin{aligned}\mathcal{A}(n,k) &= \bigcup_{x=1}^{n+1} (\mathcal{A}_x(n,k) \cup \overline{\mathcal{A}_x}(n,k)) \\ &= \bigcup_{x=1}^{\lfloor \frac{n}{2} \rfloor + 1} \left(\mathcal{A}_x(n,k) \cup \mathcal{A}_x^r(n,k) \cup \overline{\mathcal{A}_x}(n,k) \cup \overline{\mathcal{A}_x^r}(n,k) \right).\end{aligned}$$

□

Definition 6. We will call the set of all α-labelings of the $B(n,k)$ tree with $f(r) \in \{1, 2, \dots, \lfloor \frac{n}{2} \rfloor + 1\}$ the set of *principal* α-labelings and denote it by

$$\mathcal{P}(n,k) = \bigcup_{x=1}^{\lfloor \frac{n}{2} \rfloor + 1} \mathcal{A}_x(n,k).$$

Also, we will call the set of all α-labelings of the $B(n,k)$ tree with $f(r) = \lfloor \frac{n}{2} \rfloor + 1$ the set of *central* α-labelings and denote it by

$$\mathcal{C}(n,k) = \mathcal{A}_{\lfloor \frac{n}{2} \rfloor + 1}(n,k).$$

Finally, we will use the notation

$$P(n,k) = |\mathcal{P}(n,k)|$$

and

$$C(n,k) = |\mathcal{C}(n,k)|$$

for the number of principal and central α-labelings of the $B(n,k)$ tree, respectively.

Theorem 4. *For every $n, k \in \mathbb{N}$, we have*

$$\mathcal{A}(n,k) = \mathcal{P}(n,k) \cup \mathcal{P}^r(n,k) \cup \overline{\mathcal{P}}(n,k) \cup \overline{\mathcal{P}^r}(n,k),$$

where $\mathcal{P}^r(n,k) = \{f^r : f \in \mathcal{P}(n,k)\}$*,* $\overline{\mathcal{P}}(n,k) = \{\overline{f} : f \in \mathcal{P}(n,k)\}$ *and* $\overline{\mathcal{P}^r}(n,k) = \{\overline{f^r} : f \in \mathcal{P}(n,k)\}$*.*

Proof. By Corollary 3.1, we have that

$$\mathcal{A}(n,k) = \bigcup_{x=1}^{\lfloor \frac{n}{2} \rfloor + 1} \left(\mathcal{A}_x(n,k) \cup \mathcal{A}_x^r(n,k) \cup \overline{\mathcal{A}_x}(n,k) \cup \overline{\mathcal{A}_x^r}(n,k) \right).$$

Since

$$\bigcup_{x=1}^{\lfloor \frac{n}{2} \rfloor + 1} \mathcal{A}_x(n,k) = \mathcal{P}(n,k),$$

we have that

$$\mathcal{A}(n,k) = \mathcal{P}(n,k) \cup \mathcal{P}^r(n,k) \cup \overline{\mathcal{P}}(n,k) \cup \overline{\mathcal{P}^r}(n,k).$$

□

Theorem 5. *For every $n, k \geq 2$, we have*

$$A(n,k) = \begin{cases} 4P(n,k), & \text{if } n \text{ is odd} \\ 4P(n,k) - 2C(n,k), & \text{if } n \text{ is even} \end{cases}$$

Proof. We will look at the case when n is odd and the case when n is even separately.

1. If n is even, we have $n + 2 - (\lfloor \frac{n}{2} \rfloor + 1) = \lfloor \frac{n}{2} \rfloor + 1$, which means that the central α-labelings are counted twice in the partition of $\mathcal{A}(n,k)$, once in $\mathcal{P}(n,k)$ and once in $\mathcal{P}^r(n,k)$. The same argument applies to the complement of the central α-labelings, which are counted twice in $\overline{\mathcal{P}}(n,k)$ and $\overline{\mathcal{P}^r}(n,k)$. Therefore, we have
$$A(n,k) = 4P(n,k) - 2C(n,k).$$
2. If n is odd, we have $n + 2 - (\lfloor \frac{n}{2} \rfloor + 1) = \lfloor \frac{n}{2} \rfloor + 2$, which means that the central α-labelings are not twice in the partition of $\mathcal{A}(n,k)$. The same argument applies to the complement of the central α-labelings, which are not counted twice in the partition of $\mathcal{A}(n,k)$. Therefore, we have
$$A(n,k) = 4P(n,k).$$

□

Corollary 5.1. *For every $n, k \geq 2$, if $P(n,k) = 0$, then $A(n,k) = 0$, i.e. $B(n,k)$ is not α-labelable.*

Proof. If $P(n,k) = 0$ and n is odd, then we have

$$A(n,k) = 4P(n,k) = 4 \cdot 0 = 0.$$

If $P(n,k) = 0$ and n is even, then we have

$$0 \leq C(n,k) \leq P(n,k) = 0 \implies C(n,k) = 0$$

since $\mathcal{C}(n,k) \subseteq \mathcal{P}(n,k)$. Therefore,

$$A(n,k) = 4P(n,k) - 2C(n,k) = 4 \cdot 0 - 2 \cdot 0 = 0.$$

□

The $B(n,1)$ tree is isomorphic to the complete bipartite graph $K_{1,n}$. It is known that complete bipartite graphs K_{mn} are graceful [3]. In [6, Theorem 6] it was shown that every graph K_{mn} is α-labelable, therefore $B(n,1)$ tree is α-labelable. Here we give the list of all α-labelings of $B(n,1)$ tree.

Theorem 6. *The $B(n,1)$ tree is α-labelable for all $n \in \mathbb{N}$. For $n = 1$ there exists a unique α-labeling. For $n \geq 2$ there exist exactly 2 distinct α-labelings.*

Proof. The vertices of $B(n,1)$ tree are labeled with elements from $[n+1]$. The $B(1,1)$ tree has a unique labeling which is an α-labeling with $\alpha = 1$.

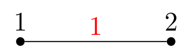


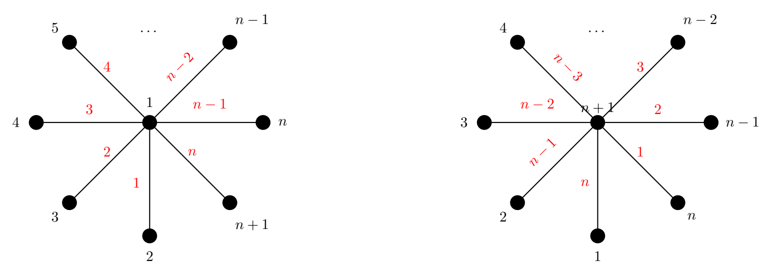
Figure 4: The unique α-labeling of $B(1,1)$

Let $n > 1$ and let f be a graceful labeling of a $B(n,1)$ tree. Assume that $f(r) = x \in \{2,3,\ldots,n\}$. Then, there must exist vertices a_i and $a_{i'}$ such that $f(a_i) = x - 1$ and $f(a_{i'}) = x + 1$. We compute

$$f^*(ra_i) = |x - (x-1)| = 1 \quad \text{ and } \quad f^*(ra_{i'}) = |x - (x+1)| = 1,$$

which is a contradiction since f is graceful. Therefore, $f(r) \in \{1, n+1\}$. In the following table we give the two unique α-labelings f_1 and f_2 of the $B(n,1)$ tree. Note that the labels of leaves a_i are uniquely determined up to isomorphism, therefore there are no other α-labelings of the $B(n,1)$ tree.

	f_1	$f_2 = f_1^r$
$f_j(r)$	1	$n+1$
$f_j(a_i)$	$i+1$	i
α_j	1	n



α-labeling f_1 of $B(n,1)$ α-labeling f_2 of $B(n,1)$

Figure 5: Two α-labelings of $B(n,1)$

The labeling f_1 is graceful because $f_1^*(ra_i) = |1 - (i+1)| = i$ for all $i \in [n]$ and therefore $\mathcal{F}_1^* = [n]$. We can see that $\alpha_1 = 1$ since $f_1(r) = 1$ and $f_1(a_1) = 2$. We can also see that $f_1(r) = 1 \leq 1 = \alpha_1 < i + 1 = f_1(a_i)$ for all $i \in [n]$. Therefore, f_1 is indeed an α-labeling of the $B(n,1)$ tree.

The labeling f_2 is the complementary labeling of f_1 and therefore it is also an α-labeling of the $B(n,1)$ tree, with $\alpha_2 = \overline{\alpha_1} = n+1-\alpha_1 = n$. $\square$

We note that $B(1,2) \cong P_3$ and $B(1,3) \cong P_4$. It was already shown in [6] that paths P_n are α-labelable. Here we give all their α-labelings.

Theorem 7. *The $B(1,2)$ tree and the $B(1,3)$ tree are have exactly 2 distinct α-labelings.*

Proof. There are 3 non-isomorphic labelings of the $B(1,2)$ tree. The labeling f_1 of $B(1,2)$ is shown in Figure 6. It is an α-labeling with $\alpha_1 = 1$. The labeling f_2 of $B(1,2)$ is the complementary labeling of f_1, so it is also an α-labeling with $\alpha_2 = \overline{\alpha_1} = 3-1 = 2$.

The only other non-isomorphic labeling of the $B(1,2)$ tree is when the central vertex is labeled by 2, however it is not even graceful.

α-labeling f_1 of $B(1,2)$ α-labeling f_2 of $B(1,2)$

Figure 6: Two α-labelings of $B(1,2)$

There are 12 non-isomorphic labelings of the $B(1,3)$ tree. Only two of them, presented in Figure 7, are α-labelings. In the case of f_1, it is easy to see $\alpha_1 = 2$. The other labeling f_2 is the complementary labeling of f_1, with $\alpha_2 = \overline{\alpha_1} = 4-2 = 2$.

α-labeling f_1 of $B(1,3)$ α-labeling f_2 of $B(1,3)$

Figure 7: Two α-labelings of $B(1,3)$

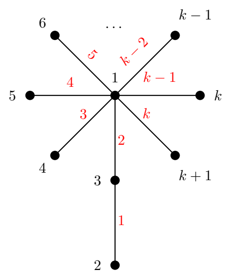
$\square$

Theorem 8. *The $B(1,k)$ tree is α-labelable for $k \geq 4$ and has exactly 4 distinct α-labelings.*

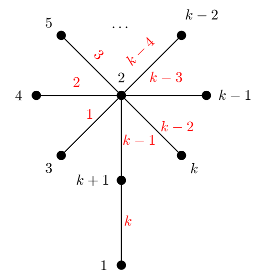
Proof. By Theorem 4, we know that we only need to find all the α-labelings of the $B(1,k)$ tree when $f(r) = 1$. The four possible labelings for $B(1,k)$ are shown in the following table. Note that the values of $f(c_1)$, $f(r)$ and $f(a_1)$ uniquely determine the values of leaves of the induced star S_k^1.

	f_1	$f_2 = f_1^r$	$f_3 = \overline{f_1}$	$f_4 = \overline{f_1^r}$
$f_i(r)$	2	1	k	$k+1$
$f_i(a_1)$	3	$k+1$	$k-1$	1
$f_i(c_1)$	1	2	$k+1$	k
$\mathcal{F}_i(L_1)$	$\{4,\ldots,k+1\}$	$\{3,\ldots,k\}$	$\{1,\ldots,k-2\}$	$\{2,\ldots,k-1\}$
α_i	2	2	$k-1$	$k-1$

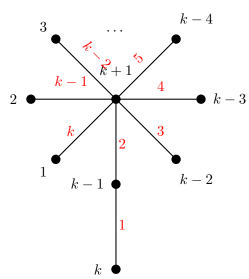
In the Figure 8 we compute the values of f^* to show that the four labelings are in fact α-labelings. We note that $\alpha_1 = 1$ for f_1 since $f_1(r) = 2$, $f(a_1) = 3$ and $f_1^*(ra_1) = |2-3| = 1$. The labeling f_2 is the reverse of f_1 so it is also an α-labeling of the $B(1,k)$ tree with $\alpha_2 = \alpha_1^r = 2$. The labeling f_3 is the complementary labeling of f_1 so it is also an α-labeling of the $B(1,k)$ tree with $\alpha_3 = \overline{\alpha_1} = k-1$. The labeling f_4 is the reverse of f_3 so it is also an α-labeling of the $B(1,k)$ tree with $\alpha_4 = \alpha_3^r = k-1$.



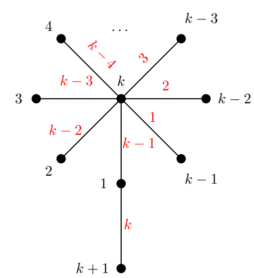
α-labeling f_1 with $\alpha_1 = 2$



α-labeling f_2 with $\alpha_2 = 2$



α-labeling f_3 with $\alpha_3 = k-1$



α-labeling f_4 with $\alpha_4 = k-1$

Figure 8: Four possible α-labelings of $B(1,k)$

We only need to show that the value of $f(a_1)$ is uniquely determined when $f(r) = 1$ and $f(c_1) = 2$. Suppose $f(a_1) \neq k+1$, i.e. $3 \leq f(a_1) \leq k$. Then, we would have $f^*(ra_1) = f(a_1) - 1 \leq k-1$ and $f(c_1a_1) = f(a_1) - 2 \leq k-2$. Now, we need to have $f^*(c_1v) = k$ for some $v \in \mathcal{L}_1$ for the labeling to be graceful. But this is not possible since $f(c_1) = 2$ and $f(v) \leq k+1$, so we would have $f^*(c_1v) = f(v) - 2 \leq k-1$. This is a contradiction, so we must have $f(a_1) = k+1$. $\square$

Theorem 9.

$$\mathcal{A}_1(2,k) = \begin{cases} \emptyset, & k = 2 \\ \{f\}, & k \geq 3 \end{cases}$$

where

$$\begin{aligned} f(r) &= 1 \\ f(c_i) &= i+1 \\ f(a_i) &= (3-i)k+1 \\ \mathcal{F}(\mathcal{L}_1) &= \{k+3, k+4, \ldots, 2k\} \\ \mathcal{F}(\mathcal{L}_2) &= \{4, 5, \ldots, k-1, k, k+2\} \end{aligned}$$

Proof. Let $k = 2$. Then, by definition, a putative α-labeling $f \in \mathcal{A}_1(2,2)$ must have $f(r) = 1$ and without loss of generality, we can set $f(c_1) = 2$, $f(c_3) = 3$. We have two cases: $f(a_1) = 4$ and $f(a_1) = 5$. If $f(a_1) = 4$, we have $f(a_2) = 5$, but then $f^*(a_1c_1) = 4 - 2 = 2 = 5 - 3 = f^*(a_2c_2)$ so f is not even graceful. If we have $f(a_1) = 5$, then we have $f(a_1c_1) = 5 - 2 = 3 = 4 - 1 = f^*(ra_2)$ so f is not even graceful. Therefore, $\mathcal{A}_1(2,2) = \emptyset$.

Let $k \geq 3$. Then, by definition, a putative α-labeling $f \in \mathcal{A}_1(2,2)$ must have $f(r) = 1$ and without loss of generality, we can set $f(c_1) = 2$, $f(c_3) = 3$.

We know that we must have $f^*(vw) = 2k$ for some $vw \in E(B(2,k))$. This is only possible for ra_1 and ra_2.

Suppose $f^*(ra_2) = 2k$, i.e. $f(a_2) = 2k+1$ and $f^*(a_2c_2) = 2k-2$. Then, we would need to have $f^*(vw) = 2k-1$. This can't be ra_1, since if $f^*(ra_1) = 2k-1$, then we would have $f(a_1) = 2k$ and $f^*(a_1c_1) = 2k-2$, but then f wouldn't even be graceful. If $f^*(vc_1) = 2k-1$, then we would have $f(v) = 2k+1$, which is not possible. If $f^*(vc_2) = 2k-1$, then we would have $f(v) = 2k+2$, but that is also not possible. Therefore, we must have $f(a_1) = 2k+1$.

Then, we must have $f^*(vc_i) = 2k-2$ for some $i \in \{1,2\}$. If, $i = 2$, we have $f(v) = f(c_2) + 2k - 2 = 2k+1$, but that is not possible since $f(a_1) = 2k+1$. Inductively, we can show that $\mathcal{F}(\mathcal{L}_1) = \{2k, 2k-1, \ldots, k+3\}$.

Now we need to have $f^*(ra_2) = k$ or $f^*(a_2c_2) = k$. If we have $f^*(a_2c_2) = k$, we would have $f(a_2) = k+3$, but $k+3 \in \mathcal{F}(\mathcal{L}_1)$, so we must have $f^*(ra_2) = k$, i.e. $f(a_2) = k+2$. The labes that we haven't yet assigned to any vertex are $\{4, 5, \ldots, k-1, k, k+2\}$, so we must have $\mathcal{F}(\mathcal{L}_2) = \{4, 5, \ldots, k-1, k, k+2\}$. This is the only possible labeling of the $B(2,k)$ tree with $f(r) = 1$, and it is exactly the labeling f. □

Theorem 10. *For any $k \geq 2$, we have*

$$\mathcal{A}_2(2,k) = \{f_y \mid k+3 \leq y \leq 2k+1\},$$

where

$$\begin{aligned}
f_y(r) &= 2 \\
f_y(c_1) &= 1 \\
f_y(c_2) &= 3 \\
f_y(a_1) &= y \\
f_y(a_2) &= 2k+5-y \\
\mathcal{F}_y(\mathcal{L}_1) &= \{y+1, y+2, \ldots, 2k+1\} \cup \{y-2i \mid 1 \leq i \leq y-k-3\} \\
\mathcal{F}_y(\mathcal{L}_2) &= (\{4, 5, \ldots, 2k+4-y\} \cup \{2k+5-y+2i \mid 1 \leq i \leq y-k-3\})
\end{aligned}$$

Proof. Firstly, we will show that $\{f_y \mid k+3 \leq y \leq 2k+1\} \subseteq \mathcal{A}_2(2,k)$, i.e. that all the f_y are α-labelings of $B(2,k)$. Then we will show that $\mathcal{A}_2(2,k) \subseteq \{f_y \mid k+3 \leq y \leq 2k+1\}$, i.e. that all the α-labelings of $B(2,k)$ with $f(r) = 2$ are exactly f_y for $k+3 \leq y \leq 2k+1$.

1. We need to show $\mathcal{F} = [2n+1]$. We can see that $\mathcal{F}(RC) = [3]$. Therefore, we need to show
$$\mathcal{F}(\mathcal{B}_1 \cup \mathcal{B}_2) = \{4, 5, \ldots, 2k+1\}$$
   We know that
$$\begin{aligned}
\mathcal{F}(\mathcal{B}_1) &= \{y+1, \ldots, 2k+1\} \cup \{y-2(y-k-3), y-2(y-k-4), \ldots, y\} \\
&= \{y+1, \ldots, 2k+1\} \cup \{2k+6-y, 2k+8-y, \ldots, y\}
\end{aligned}$$

and

$$\begin{aligned}\mathcal{F}(\mathcal{B}_2) &= \{4,\ldots,2k+5-y\} \cup \{2k+5-y+2,\ldots,2k+5-y+2(y-k-3)\}\\ &= \{4,5,\ldots,2k+5-y\} \cup \{2k+7-y, 2k+9-y,\ldots,y-1\}\end{aligned}$$

Now, we can see that

$$\begin{aligned}\mathcal{F}\left(\mathcal{B}_1 \cup \mathcal{B}_2\right) &= \mathcal{F}(\mathcal{B}_1) \cup \mathcal{F}(\mathcal{B}_2)\\ &= \{y+1,\ldots,2k+1\} \cup \{2k+6-y, 2k+8-y,\ldots,y\}\\ &\quad \cup \{4,5,\ldots,2k+5-y\} \cup \{2k+7-y, 2k+9-y,\ldots,y-1\}\\ &= \{4,5,\ldots,2k+5-y\} \cup \{2k+6-y,\ldots,y\} \cup \{y+1,\ldots,2k+1\}\\ &= \{4,5,\ldots,2k+1\}\end{aligned}$$

Secondly, we need to show $\mathcal{F}^* = [2k]$.
We know that

$$\begin{aligned}\mathcal{F}^*(\mathcal{B}_1) &= \{y+1-1,\ldots,2k+1-1\} \cup \{2k+6-y-1,\ldots,y-1\}\\ &= \{y,\ldots,2k\} \cup \{2k+5-y, 2k+7-y,\ldots,y-1\},\end{aligned}$$

$$\begin{aligned}\mathcal{F}^*(\mathcal{B}_2) &= \{4-3, 5-3,\ldots,2k+5-y-3\} \cup \{2k+7-y-3,\ldots,y-1-3\}\\ &= \{1,2,\ldots,2k+2-y\} \cup \{2k+4-y, 2k+6-y,\ldots,y-4\}\end{aligned}$$

and also, $f^*(ra_1) = y-2$ and $f^*(ra_1) = 2k+3-y$. Thus, we have

$$\begin{aligned}\mathcal{F}^* &= \{y,\ldots,2k\} \cup \{2k+5-y,\ldots,y-1\} \cup \{1,2,\ldots,2k+2-y\}\\ &\quad \cup \{2k+4-y, 2k+6-y,\ldots,y-4\} \cup \{y-2, 2k+3-y\}\\ &= \{y,\ldots,2k\} \cup \{2k+3-y,\ldots,y-1\} \cup \{1,2,\ldots,2k+2-y\}\\ &\quad \cup \{2k+4-y, 2k+6-y,\ldots,y-2\}\\ &= \{y,\ldots,2k\} \cup \{2k+3-y, 2k+4-y\ldots,y-1\} \cup \{1,2,\ldots,2k+2-y\}\\ &= \{1,2,\ldots,2k\} = [2k]\end{aligned}$$

Finally, we know that the α inequality holds since $f(c_i) \leq 3 < f(v)$ for all $v \in \mathcal{B}_i$ and $i = 1,2$, as well as $f(r) = 2 \leq 3 < f(a_i)$ for $i = 1,2$. Thus, we have shown $\{f_y \mid k+3 \leq y \leq 2k+1\} \subseteq \mathcal{A}_2(2,k)$.

2. Let $f \in \mathcal{A}_2(2,k)$ be an α-labeling of $B(2,k)$ with $f(r)$. Then we know that, without loss of generality, we have $f(c_1) = 1$ and $f(c_2) = 3$, and we also know that $4 \in \mathcal{F}(\mathcal{B}_2)$ since $\alpha = 3$ and $2k+1 \in \mathcal{F}(\mathcal{B}_1)$ since we need $2k \in \mathcal{F}^*$.

   Let $f(a_1) = x$ for some $x \in \{5,\ldots,2k\}$. Then we know $f(ra_1) = x-2$ and $f^*(a_1c_1) = x-1$. Now, we know that $f(v) \neq x+1$ for all $v \in \mathcal{B}_2$, since if it were, we would have $f^*(vc_2) = x+1-3 = x-2$, but then the labeling wouldn't even be graceful. Furthermore, we can inductively reason that $\{x, x+1,\ldots,2k+1\} \subseteq \mathcal{F}(\mathcal{B}_1)$. This means that

$$\begin{aligned}|\{x, x+1,\ldots 2k+1\}| &\leq |\mathcal{F}(\mathcal{B}_1)|\\ 2k+1-x+1 &\leq k-1\\ 2k+3-x &\leq k\\ x &\geq k+3\end{aligned}$$

Now we have $\{x-1, x, x+1, \ldots, 2k\} \subseteq \mathcal{F}^*(\mathcal{B}_1)$ and $f^*(ra_1) = x-2$. We also have $x-1 \notin \mathcal{F}(\mathcal{B}_1)$, since we have $f^*(ra_1) = x-2$ and f wouldn't even be graceful. Therefore, $x-1 \in \mathcal{F}(\mathcal{B}_2)$.

Let $f(a_2) = z$. Then we have $f^*(ra_2) = z-2$ and $f^*(a_2c_2) = z-3$. Now we know that $f(v) \neq z-1$ for all $v \in \mathcal{B}_1$, since if it were, we would have $f^*(vc_1) = z-1-1 = z-2$, but then the labeling wouldn't even be graceful. Furthermore, we can inductively reason that $\{z, z-1, \ldots, 4\} \subseteq \mathcal{F}(\mathcal{B}_1)$. This means that

$$\begin{aligned} |\{z, z-1, \ldots, 4\}| &\leq |\mathcal{F}(\mathcal{B}_1)| \\ z-4+1 &\leq k-1 \\ z &\leq k+2 \end{aligned}$$

Now we have $\{z-3, z-4, \ldots, 1\} \subseteq \mathcal{F}(\mathcal{B}_2)$. Furthermore, we know that $z+1 \in \mathcal{F}(\mathcal{B}_1)$ since, otherwise we would have $f^*(vc_2) = z+1-3 = z-2 = f^*(ra_2)$ for some $v \in \mathcal{B}_2$ and the labeling wouldn't even be graceful. Now, we know that $z+2 \in \mathcal{F}(\mathcal{B}_2)$ because we need to have $z-1 \in \mathcal{F}^*$. Furthermore, we can inductively reason that $\{z+1, z+3, \ldots, z+2t-1\} \subseteq \mathcal{F}(\mathcal{B}_1)$ for some $t \geq 0$ as well as $\{z+2, z+4, \ldots, z+2u\} \subseteq \mathcal{F}(\mathcal{B}_2)$ for some $u \geq 0$. Then, we have $\mathcal{F}(\mathcal{B}_1) = \{x, x+1, \ldots, 2k+1\} \cup \{z+1, z+3, z+2t-1\}$, i.e.

$$\begin{aligned} k-1 &= |\{x, x+1, \ldots, 2k+1\}| + |\{z+1, z+3, z+2t-1\}| \\ k-1 &= (2k+1-x+1)+t \\ 0 &= k+3-x+t \\ t &= x-k-3 \end{aligned}$$

Let's look at the vertex v with the label $x-1$. We know it can't be in $\mathcal{B}_1$ since then we would have $f^*(vc_1) = x-2 = f^*(ra_1)$. Therefore, we must have $x-1 \in \mathcal{F}(\mathcal{B}_2)$. Now we need to find the edge with the induced label $x-3$. This can only be wc_1 for some $w \in \mathcal{B}_1$. Furthermore, we can inductively reason that $\{x-1, x-3, \ldots, x-2t'+1\} \in \mathcal{F}(\mathcal{B}_2)$ and $\{x-2, x-4, \ldots, x-2u'\} \in \mathcal{F}(\mathcal{B}_1)$ for sume $t', u' \geq 0$. We now know

$$\begin{aligned} z+2t-1 &= x-2 \\ z+2(x-k-3)+1 &= x \\ z &= 2k+5-x \end{aligned}$$

since $\{z+1, z+3, z+2t-1\} = \{x-2, x-4, \ldots, x-2u'\}$. Thus we have that f is exactly f_x from the set $\{f_y \mid k+3 \leq y \leq 2k+1\}$, i.e. $\mathcal{A}_2(2,k) \subseteq \{f_y \mid k+3 \leq y \leq 2k+1\}$.

Thus, we have $\mathcal{A}_2(2,k) = \{f_y \mid k+3 \leq y \leq 2k+1\}$. □

Corollary 10.1.

$$A(2,k) = \begin{cases} 2, & k = 1,2 \\ 2(k+1), & k \geq 3 \end{cases}$$

Proof. We will look at three different cases: $k=1$, $k=2$ and $k \geq 3$.

1. Let $k=1$. By Theorem 6, we know that $A(2,1) = 2$.

2. Let $k = 2$. From the Theorems 9 and 10, we can see that $P(2,2) = A_1(2,2) + A_2(2,2) = 0 + (2 \cdot 2 + 1 - (2+3) + 1) = 1$ and $C(2,2) = A_2(2,2) = 1$. Therefore, by Theorem 5 we have that $A(2,2) = 4P(2,2) - 2C(2,2) = 4 - 2 = 2$.
3. Let $k \geq 3$. From the Theorems 9 and 10, we can see that $P(2,k) = A_1(2,k) + A_2(2,k) = 1 + (2k + 1 - (k+3) + 1) = 1 + k - 1 = k$ and $C(2,k) = A_2(2,k) = (2k + 1 - (k+3) + 1) = k - 1$. Therefore, by Theorem 5 we have that $A(2,k) = 4P(2,k) - 2C(2,k) = 4k - 2(k-1) = 2k + 2 = 2(k+1)$.

□

The construction in the following theorem is inspired by the construction of the graceful labeling of superstar trees in [4]. Our result differs from theirs in the fact that we proved that this labeling is an α-labeling, while in [4] it is only proven that it is a graceful labeling. Also, we are only looking at banana trees, not superstar trees. The labeling given by this construction is also the reverse labeling of the labeling given by the construction in [9].

Theorem 11. *The $B(n,k)$ tree is α-labelable for all $n \geq 3$ and $k > n$.*

Proof. The $B(n,k)$ tree has $[nk+1]$ vertices. We define a labeling $f : V(B(n,k)) \to \mathbb{N}$ (Figure 9) as follows:

$$\begin{array}{lcl} f(r) & = & n+1 \\ f(c_i) & = & i \\ f(a_i) & = & nk + (1-k)i + (n+2-i) \\ \mathcal{F}(\mathcal{L}_i) & = & \{nk + (1-k)i + 2, \dots, nk + (1-k)i + k\} \setminus \{f(a_i)\}, \end{array}$$

where $i = 1, \dots, n$.

First, we show that each vertex of the $B(n,k)$ tree is labeled by the distinct element of $[nk+1]$. Obviously, vertices $c_1, \dots, c_n, r$ are labeled by a distinct elements of $[n+1]$. Now we discuss the lables of all other vertices, that is, we deterimine $\mathcal{F}(\mathcal{B}) = \mathcal{F}(\mathcal{B}_1) \cup \dots \cup \mathcal{F}(\mathcal{B}_n)$, where $\mathcal{B}_i = \{a_i\} \cup \mathcal{L}_i = \{a_i, l_{i,1}, \dots, l_{i,k-2}\}$. Note that for the vertex a_i, $i \in [n]$, it holds that

$$nk + (1-k)i + 2 - f(a_i) = i - n - 2 \leq 0$$

and

$$nk + (1-k)i + k - f(a_i) = k - n - 2 + i \geq 0,$$

since $k > n$. Therefore, the vertices of $\mathcal{B}_i$ are labeled by the distinct elements of

$$\mathcal{F}(\mathcal{B}_i) = \{nk + (1-k)i + 2, \dots, nk + (1-k)i + k\}.$$

We also have that $\mathcal{F}(\mathcal{B}_i) \cap \mathcal{F}(\mathcal{B}_{i+1}) = \emptyset$ since

$$\mathcal{F}(\mathcal{B}_{i+1}) = \{nk + (1-k)i + 3 - k, \dots, nk + (1-k)i + 1\} \tag{1}$$

and

$$\min \mathcal{F}(\mathcal{B}_i) - \max \mathcal{F}(\mathcal{B}_{i+1}) = (nk + (1-k)i + 2) - (nk + (1-k)i + 1) = 1.$$

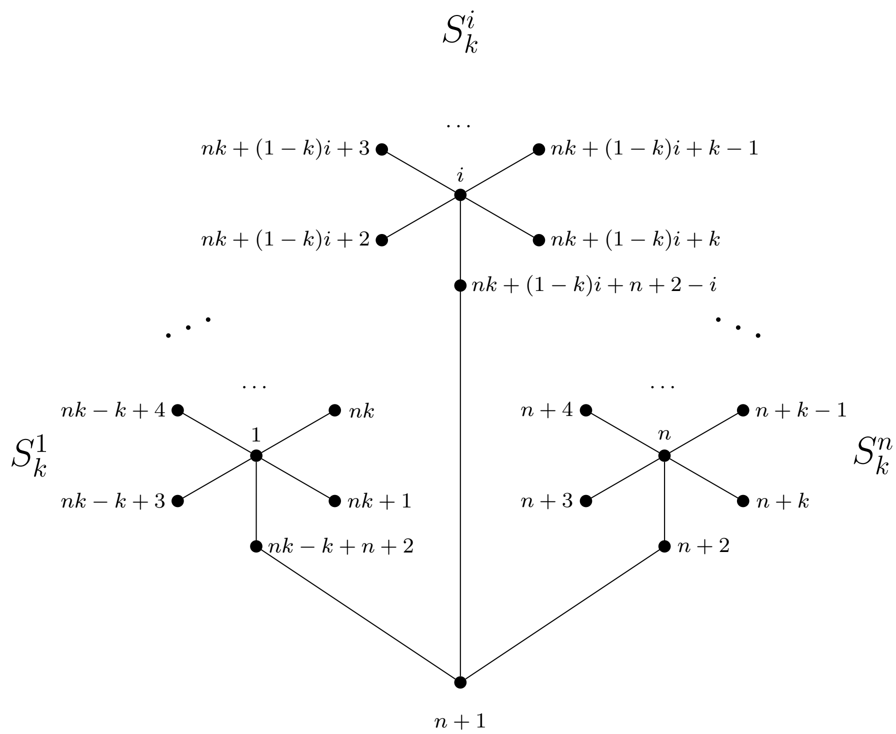


Figure 9: The labeling f of the $B(n,k)$ tree

Therefore,

$$\begin{aligned}\mathcal{F}(\mathcal{B}) &= \mathcal{F}(\mathcal{B}_n)\cup\cdots\cup\mathcal{F}(\mathcal{B}_1) = \\ &= \{n+2,\ldots n+k\}\cup\cdots\cup\{nk-k+3,\ldots,nk+1\} = \\ &= \{n+2,n+3,\ldots,nk+1\},\end{aligned}\tag{2}$$

and we conclude that the vertices of the $B(n,k)$ tree are labeled with distinct elements of $[nk+1]$.

Now we show that the labeling f is graceful. The edges of S_k^i, $i\in[n]$, are $E(S_k^i) = \{c_ia_i,\, c_il_{i,1},\ldots,c_il_{i,k-2}\}$. Since $f(c_i)=i$ and the vertices of $\mathcal{B}_i$ are labeled by distinct elements of (1) we have that

$$\mathcal{F}^*(E(S_k^i)) = \{f^*(c_iv)=|f(c_i)-f(v)|\,:\,v\in\mathcal{B}_i\} = \{nk-ik+2,\ldots,nk-ik+k\}.$$

The remaining edges of the $B(n,k)$ tree are $ra_i, i\in[n]$, labeled by $f^*(ra_i) = |n+1-(nk+(1-k)i+(n+2-i))| = nk-ik+1$. Therefore, the edges of one branch of the $B(n,k)$ tree are labeled by

$$\{nk-ik+1,\ldots,nk-ik+k\}$$

and the edges of the $B(n,k)$ tree are labeled by

$$\bigcup_{i=1}^{n}\{nk-ik+j\,:\,j\in[k]\} = \{nk-ik+j\,:\,i\in[n],j\in[k]\} = [nk].$$

Lastly, we show that f is an α-labeling. We can see that $\alpha = n+1$ since $f^*(ra_n) = |n+1-(n+2)| = 1$. Following (2) we have that $f(v) \geq \alpha$, therefore for every edge $c_i v$, $i \in [n]$, $v \in \mathcal{B}_i$, it holds that $f(c_i) = i\alpha < f(v)$. Finally, for the edges ra_i, $i \in [n]$, we have $f(r) = n+1 \leq \alpha < n+2 < n+2+(n-i)k = f(a_i)$ Therefore, we conclude that f is an α-labeling of the $B(n,k)$ tree. □

Theorem 12. *The $B(n,k)$ tree is α-labelable for all $n \in \mathbb{N}$ and $\frac{n}{2}+1 \leq k \leq n+1$.*

Proof. Let $f : V(B(n,k)) \to \mathbb{N}$ be a labeling of the $B(n,k)$ tree with $[nk+1]$ vertices, defined as follows (Figure 10):

$$f(r) = n+2-k$$
$$f(c_i) = \begin{cases} n+2-i, & i < k \\ n+1-i, & i \geq k \end{cases}$$
$$f(a_i) = \begin{cases} n+2-k+ki, & i < k \\ n+3-2k+ki, & i \geq k \end{cases}$$
$$\mathcal{F}(\mathcal{L}_i) = \{n+3-k+(k-1)i, \ldots, n+1+(k-1)i\} \setminus \{f(a_i)\}$$

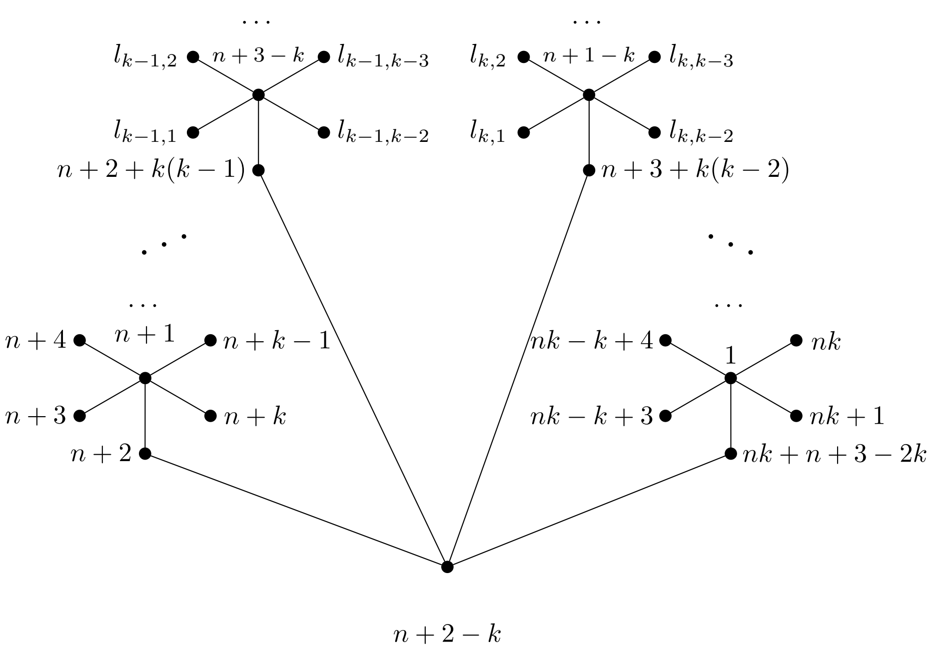


Figure 10: The labeling f of the $B(n,k)$ tree

First, we show that each vertex of the $B(n,k)$ tree is labeled by a distinct element of $[nk+1]$. Obviously, vertices $c_n, \ldots, c_k, r, c_1, \ldots, c_{k-1}$ are labeled by the distinct elements of $[n+1]$. Now we discuss the labels of all other vertices $v \in \mathcal{B}$. For $1 \le i < k$, the vertex a_i is labeled by $f(a_i) = n+2-k+ki$. We can see that then

$$n+3-k+(k-1)i-f(a_i) = 1-i \le 0$$

and

$$n+1+(k-1)i-f(a_i) = k-1-i \ge 0.$$

For $k \le i \le n$, that is $k \le i \le 2k-2$ since we have $k \le \frac{n}{2}+1$, the vertex a_i is labeled by $f(a_i) = n+3-2k+ki$. We can see that then

$$n+3-k+(k-1)i-f(a_i) = k-i \le 0$$

and

$$n+1+(k-1)i-f(a_i) = 2k-2i \ge 0.$$

Therefore, for $i \in [n]$, the vertices of $\mathcal{B}_i$ are labeled by the distinct elements of

$$\mathcal{F}(\mathcal{B}_i) = \{n+3-k+(k-1)i, \dots, n+1+(k-1)i\}.$$

We also have that $\mathcal{F}(\mathcal{B}_i) \cap \mathcal{F}(\mathcal{B}_{i+1}) = \emptyset$ since

$$\mathcal{F}(\mathcal{B}_{i+1}) = \{n+2+(k-1)i, \dots, n+k+(k-1)i\} \tag{3}$$

and thus we have

$$\min \mathcal{F}(\mathcal{B}_{i+1}) - \max \mathcal{F}(\mathcal{B}_i) = n+2+(k-1)i-(n+1+(k-1)i) = 1.$$

Therefore,

$$\begin{aligned} \mathcal{F}(\mathcal{B}) &= \mathcal{F}(\mathcal{B}_1) \cup \dots \cup \mathcal{F}(\mathcal{B}_n) = \\ &= \{n+2, \dots n+k\} \cup \dots \cup \{nk-k+3, \dots, nk+1\} = \\ &= \{n+2, n+3, \dots, nk+1\}, \end{aligned} \tag{4}$$

and we conclude that the vertices of the $B(n,k)$ tree are labeled with distinct elements of $[nk+1]$.

Now we show that the labeling f is graceful. The vertices of $\mathcal{B}_i$ are labeled by distinct elements of (4). We will look at 2 cases.

1. If $i < k$, then $f(c_i) = n+2-i$ and we have that

   $$\mathcal{F}^*(E(S_k^i)) = \{f^*(c_i v) = |f(c_i) - f(v)| \,:\, v \in \mathcal{B}_i\} = \{ki+1-k, \dots, ki-1\}.$$

   In addition, $f^*(ra_i) = ki$, so the edges of one branch of the $B(n,k)$ tree are labeled by the distinct elements of

   $$\{ki+1-k, \dots, ki\}.$$

   So, we have that the edges of all branches for $i < k$ are labeled by the distinct elements of

   $$\{1, 2, \dots, k(k-1)\}.$$

2. If $i \ge k$, then $f(c_i) = n+1-i$, and we have

   $$\mathcal{F}^*(E(S_i)) = \{ki+2-k, \dots, ki\}.$$

Also, then $f^*(ra_i) = k(i-1)+1$. so the edges of one branch of the $B(n,k)$ tree are labeled by the distinct elements of

$$\{ki+1-k, \dots, ki\}.$$

So, we have that the edges of all branches for $i \geq k$ are labeled by the distinct elements of

$$\{k(k-1)+1, \dots, nk\}.$$

Therefore, the edges of the $B(n,k)$ tree are labeled by the distinct elements of $[nk]$, and f is a graceful labeling.

Finally, we need show to that f is an α-labeling. We can see that $\alpha = n+1$ since $f^*(ra_n) = |n+1-(n+2)| = 1$. Following (4) we have that $f(v) \geq \alpha$, therefore for every edge $c_i v$, $i \in [n]$, $v \in \mathcal{B}_i$, it holds that $f(c_i) \leq n+1 = \alpha < f(v)$. Finally, for the edges ra_i, $i \in [n]$, we have $f(r) = n+2-k \leq \alpha < f(a_i)$. Therefore, we conclude that f is an α-labeling of the $B(n,k)$ tree. $\square$

The results of the previous two theorems can be combined as follows.

Corollary 12.1. *The $B(n,k)$ tree is α-labelable for all $n \geq 3$ and $k \geq \frac{n}{2}+1$.*

4 Non-existence results

Theorem 13. *The $B(n,k)$ tree is not α-labelable for all $n \geq 3$ and $1 < k \leq \frac{n}{2}$.*

Proof. All $B(n,k)$ trees are graceful [7, 8]. Assume there exists a graceful labeling f such that $f \in \mathcal{P}(n,k)$, i.e. f is a principal α-labeling of the $B(n,k)$ tree. There are 2 possible types of α-labelings in $\mathcal{P}(n,k)$

Since $f \in \mathcal{P}(n,k)$, we have that $f(r) \leq \lfloor \frac{n}{2} \rfloor + 1$ and, without loss of generality, we have

$$f(c_i) = \begin{cases} i, & i < f(r) \\ i+1, & i \geq f(r) \end{cases}$$

We know that $\mathcal{F}(RC) = [n+1]$ since $f \in \mathcal{P}(n,k)$ and therefore $f(a_i) \geq n+2$ for all $i \in [n]$. Thus, we also have that

$$f^*(ra_i) = f(a_i) - f(r) \geq n+2 - \left\lfloor \frac{n}{2} \right\rfloor - 1 \geq n+1 - \frac{n}{2} = \frac{n}{2} + 1 > k.$$

Thus, all the edges vw for which $f^*(vw) \leq k$, must be from $S_k^1 \cup S_k^2 \cup \dots \cup S_k^n$, i.e. must be of the form $c_i v$ where $v \in \mathcal{B}_i$. We will use the notation $v^{(x)} c_{i(x)}$ for the edge for which $f^*(v^{(x)} c_{i(x)}) = x$.

Let us examine $v^{(1)} c_{i(1)}$. Since, we know all vertices that have labels in $[n+1]$, we know that $f(v^{(1)}) \geq n+2$. Therefore, we have

$$1 = f^*(v^{(1)} c_{i(1)}) = f(v^{(1)}) - f(c_{i(1)}) \geq n+2 - f(c_{i(1)}) \Rightarrow f(c_{i(1)}) \geq n+1.$$

Therefore, $v^{(1)} \in \mathcal{B}_n$, since $f(c_n) = n+1$ and $f(c_i) \leq n$ for all $i \in [n-1]$. Now, we will show that $\mathcal{B}_n = \{v^{(1)}, v^{(2)}, \dots, v^{(k-1)}\}$, by induction.

Let us suppose that we have that $V_{m-1} = \{v^{(1)}, v^{(2)}, \dots, v^{(m-1)}\} \subseteq \mathcal{B}_n$ for some $m \in [k-2]$. Then, we know that $\mathcal{F}(V_{m-1}) = \{n+2, n+3, \dots, n+1+m-1\}$, and the smallest vertex label we haven't yet found is $n+1+m-1+1 = n+m+1$, and therefore $f(v^{(m)}) \geq n+m+1$. Furthermore, we have

$$m = f^*(v^{(m)}c_{i(m)}) = f(v^{(m)}) - f(c_{i(m)}) \geq n + m + 1 - f(c_{i(1)}) \Rightarrow f(c_{i(1)}) \geq n + 1.$$

Thus, $v^{(m)} \in \mathcal{B}_n$ since $f(c_n) = n+1$ and $f(c_i) \leq n$ for all $i \in [n-1]$. Furthermore, now we know that $\mathcal{B}_n = \{v^{(1)}, v^{(2)}, \ldots, v^{(k-1)}\}$, and $f(v^{(i)}) = n+1+i$ for all $i \in [k-1]$.

Let us now examine $v^{(k)}c_{i(k)}$. Since $\mathcal{F}(RC) = [n+1]$, and $\mathcal{F}(\mathcal{B}_n) = \{n+2, n+3, \ldots, n+k\}$, we know that $f(v^{(k)}) \geq n+k+1$, and we also know that $f(c_{i(k)}) \leq n$. Therefore, we have:

$$f^*(v^{(k)}c_{i(k)}) = f(v^{(k)}) - f(c_{i(k)}) \geq n + k + 1 - n = k + 1$$

Therefore, we don't have an edge vw for which $f^*(vw) = k$ and we conclude that f is not even a graceful labeling. Thus, we know that $P(n,k) = 0$ for every $n \geq 2$ and therefore the $B(n,k)$ tree is not α-labelable for all $n \geq 3$ and $1 < k \leq \frac{n}{2}$ by Corollary 5.1. ☐

5 Conclusion

All of our results on the α-labelability of the $B(n,k)$ trees are summarized in the next graph which shows whether an α-labeling exists for $B(n,k)$.

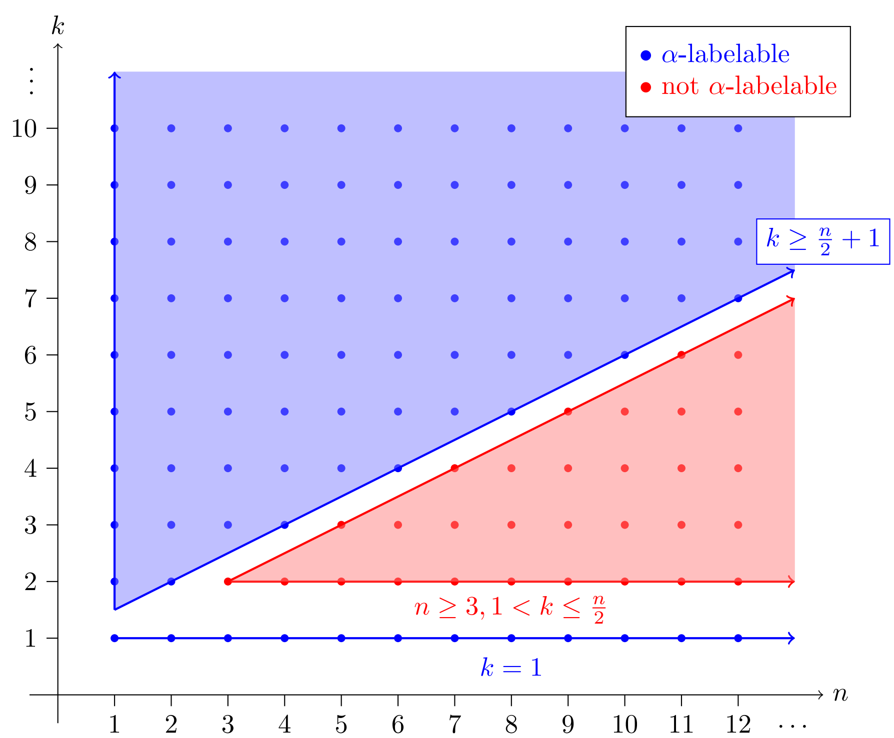


Figure 11: Existence of an α-labeling for the $B(n,k)$ tree

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